Illiquidity premium

Combined factor-model portfolio test

Practitioner factor-model runner

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Abstract

Model. Illiquidity premium, one long-short book from 2 sleeves (PriceImpact, Thinness). 5857 symbols (full screened universe), 2005-08-01 to 2026-07-01 (252 months).

Earned. -2.35% annual arithmetic return (-2.57% compounded), 7.16% volatility, Sharpe -0.33, HAC t=-1.50.

Survives? vs SPY market benchmark: alpha -2.63%, t=-1.64. vs VAL+MOM+BAB+QMJ: alpha 0.46%, t=0.34. Conventional bar is $|t| \geq 2$.

Best swap. (Standard model, PriceImpact for BAB) improves: Sharpe 0.24 against -0.06 (+0.29). **Verdict.** Reject for now: it clears neither control at |t| >= 2.

Exploratory model diagnostic from a single configured run, not evidence from an untouched confirmatory sample.

1 Research objective

Test whether the configured model earns a useful next-month return and whether the benchmark or standard factors explain it. Signals are observed at month-end; returns are next-calendar-month simple returns.

Construction. Normalize within month, combine components within sleeves, then combine sleeves at the stock level using Price Impact 50%, Thinness 50%. Construct one portfolio; apply beta neutrality and costs once. Sleeve returns are not averaged.

2 Configuration and sample

Factor	**Composite signals**	**Signal transform**	**Composite transform**	**Winsor**	**Min inputs**	**Mean inputs**
Price Impact	amihud_illiq_21d	rank	rank	1%-99%	1	1.0
Thinness	dollar_volume_21d	rank	rank	1%-99%	1	1.0

Model construction	**Book**	**Beta neutral**	**Bucket frac.**	**Cost bps**	**Min factors**	**Benchmark**
signal weighted	long-short	no	0.33	0.0	2	SPY

Portfolio	**Mean stocks**	**Minimum stocks**
Combined model	1915.1	20
Price Impact	1915.1	20
Thinness	1931.0	21

Sample: 5,857 symbols, 494,433 stock-months, 2001-07-01–2026-08-01. Screens: price 1.0, 21-day dollar volume None, market cap None, top-cap fraction 0.75, exclude financials False. Benchmark: SPY. Factor controls: standard_recreated_factors.csv.

3 Portfolio returns

Raw net performance on one shared window. Return and volatility use monthly arithmetic moments; t is HAC.

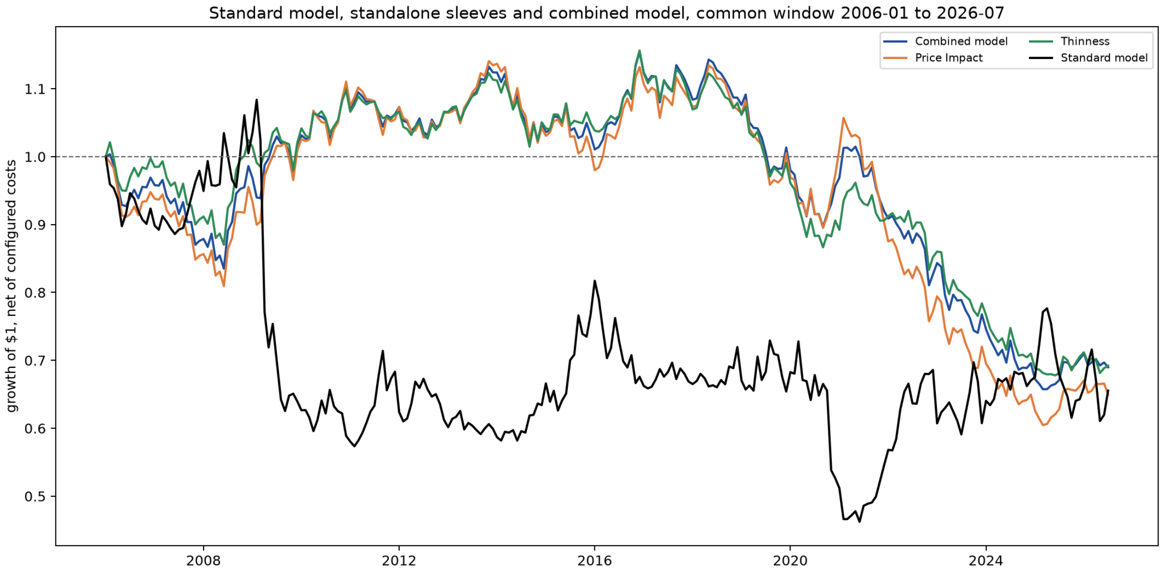

Figure 1: Growth of one dollar in each book, net of configured costs, on the common window. The path begins at one before the first return month.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	252	-2.35%	-2.57%	7.16%	-0.33	-1.50
Price Impact	252	-2.51%	-2.78%	7.89%	-0.32	-1.41
Thinness	252	-2.45%	-2.65%	6.81%	-0.36	-1.70
Standard model	246	-0.98%	-2.04%	14.42%	-0.07	-0.30

4 Benchmark fit and benchmark-orthogonalized returns

Regress each raw net return on SPY market benchmark. R2 is variance explained; alpha is the annualized intercept; t and F use HAC inference. This is separate from standard-factor attribution. Estimated MKT loading: 0.024. Annual raw return -2.35% = controlled alpha -2.63% + benchmark contribution 0.28%. Orthogonal means zero sample covariance with the benchmark, not a lower mean. This is an ex-post regression diagnostic, not beta-neutral construction or a tradeable improvement.

Portfolio	N	Loading (MKT)	t (loading)	R2 (SPY market benchmark)	F
Combined model	252	0.02	0.74	0.003	0.54
Price Impact	252	0.09	2.30	0.027	5.29
Thinness	252	-0.06	-1.98	0.015	3.91
Standard model	246	-0.57	-8.01	0.356	64.22

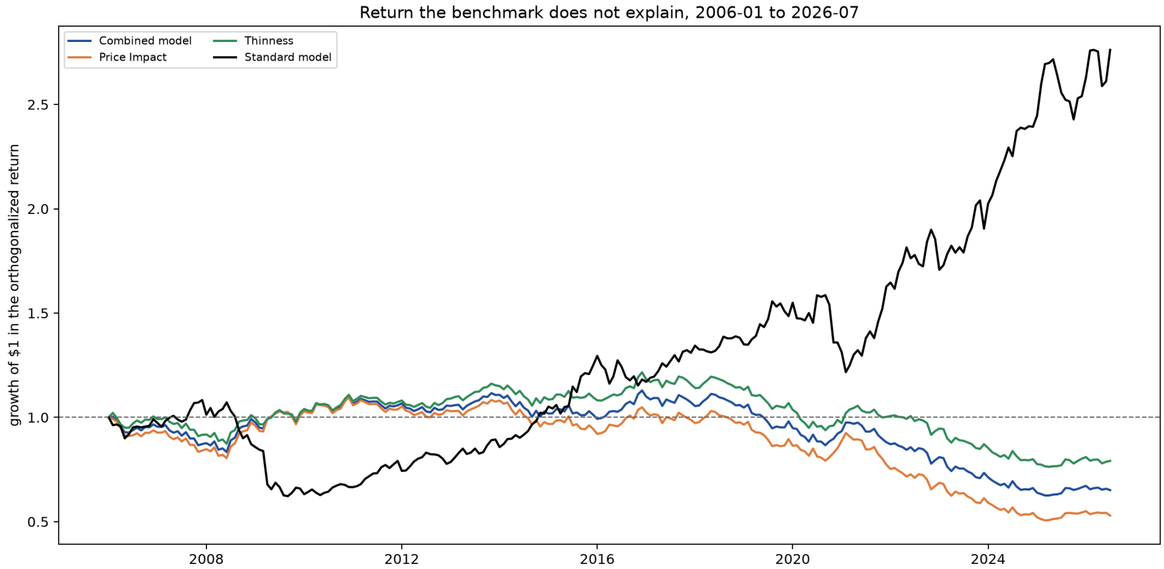

Figure 2: Returns orthogonalized to the SPY market benchmark: growth of one dollar in each book once the SPY market benchmark component is removed. A flat path earns nothing the benchmark does not already deliver.

The table summarizes the residual paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	252	-2.63%	-2.84%	7.15%	-0.37	-1.68
Price Impact	252	-3.50%	-3.74%	7.78%	-0.45	-1.99
Thinness	252	-1.81%	-2.02%	6.75%	-0.27	-1.26
Standard model	246	5.65%	5.08%	11.57%	0.49	2.11

5 Attribution to configured factor controls

Regress each raw net return simultaneously on VAL+MOM+BAB+QMJ: $r_{p,t} = \alpha + \sum_k \beta_k f_{k,t} + \varepsilon_t$. The first table gives the loadings, with HAC t in brackets.

Portfolio	VAL	MOM	BAB	QMJ
Combined model	0.02 (0.50)	-0.13 (-3.79)	-0.06 (-1.23)	-0.11 (-3.72)
Price Impact	-0.02 (-0.47)	-0.14 (-3.57)	-0.06 (-1.26)	-0.20 (-5.26)
Thinness	0.05 (1.51)	-0.12 (-3.85)	-0.04 (-0.88)	-0.01 (-0.55)

Return contribution is $\beta_k \times 12 \times \overline{f_k}$. Explained plus alpha equals total annual return.

Portfolio	VAL	MOM	BAB	QMJ	Explained	Total
Combined model	-0.02%	-0.72%	-0.17%	-0.87%	-1.78%	-1.32%
Price Impact	0.02%	-0.77%	-0.20%	-1.54%	-2.50%	-1.38%
Thinness	-0.06%	-0.69%	-0.12%	-0.11%	-0.97%	-1.57%

5.1 Simultaneous factor spanning and factor-orthogonalized returns

The same regression as a spanning test. R2 is the share of monthly net-return variance jointly explained by the factor controls.

Portfolio	N	R2 (4F)	F
Combined model	230	0.181	15.37
Price Impact	230	0.260	24.06
Thinness	230	0.111	6.42

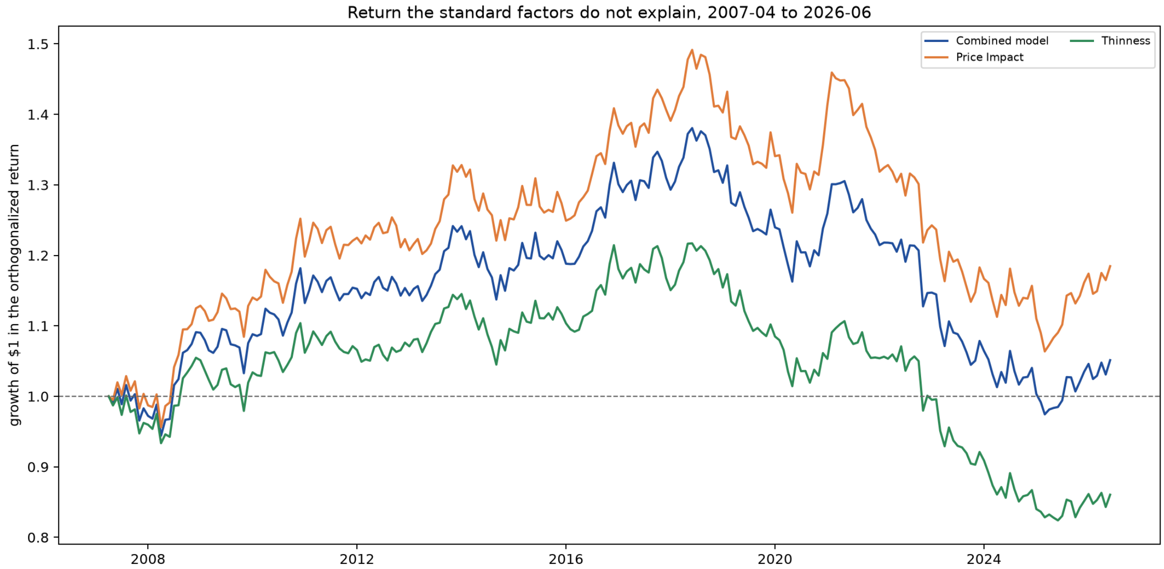

Figure 3: Returns orthogonalized to VAL+MOM+BAB+QMJ: growth of one dollar in each book once its factor exposures are removed. This is the Alpha column above, drawn as a path.

The table summarizes the factor-residual paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	230	0.46%	0.26%	6.35%	0.07	0.36
Price Impact	230	1.11%	0.89%	6.75%	0.16	0.79
Thinness	230	-0.60%	-0.78%	6.10%	-0.10	-0.50

5.2 The new model against the reference model

Regress the combined model on Standard model as one return series. High R2 with zero alpha means the reference reproduces the model.

Portfolio	N	Alpha	t	R2 (Standard model)	F
Combined model	246	-1.79%	-1.41	0.210	86.85

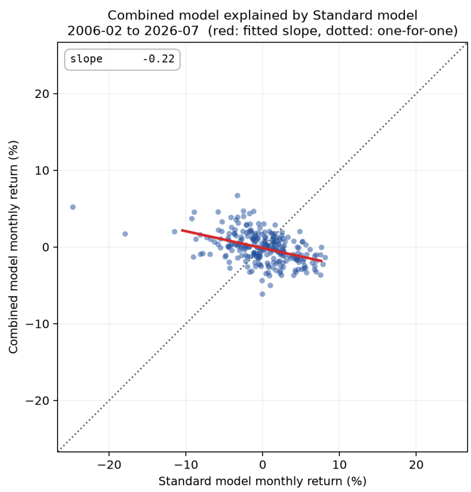

Figure 4: Each month of the combined model against Standard model, with the fitted slope. A tight cloud along the line is a model the reference reproduces; vertical spread is what it cannot.

5.3 Correlation and principal components

Check whether a sleeve duplicates a standard factor. The combined model is excluded because its overlap with its own sleeves is mechanical. PCA uses standardized sleeve and standard-factor returns over 230 common months. The share shown in the plot is the largest covariance-matrix eigenvalue divided by the sum of all eigenvalues: the fraction captured by one dominant co-movement, not return or regression R2. The combined model is excluded because its overlap with its own sleeves is mechanical.

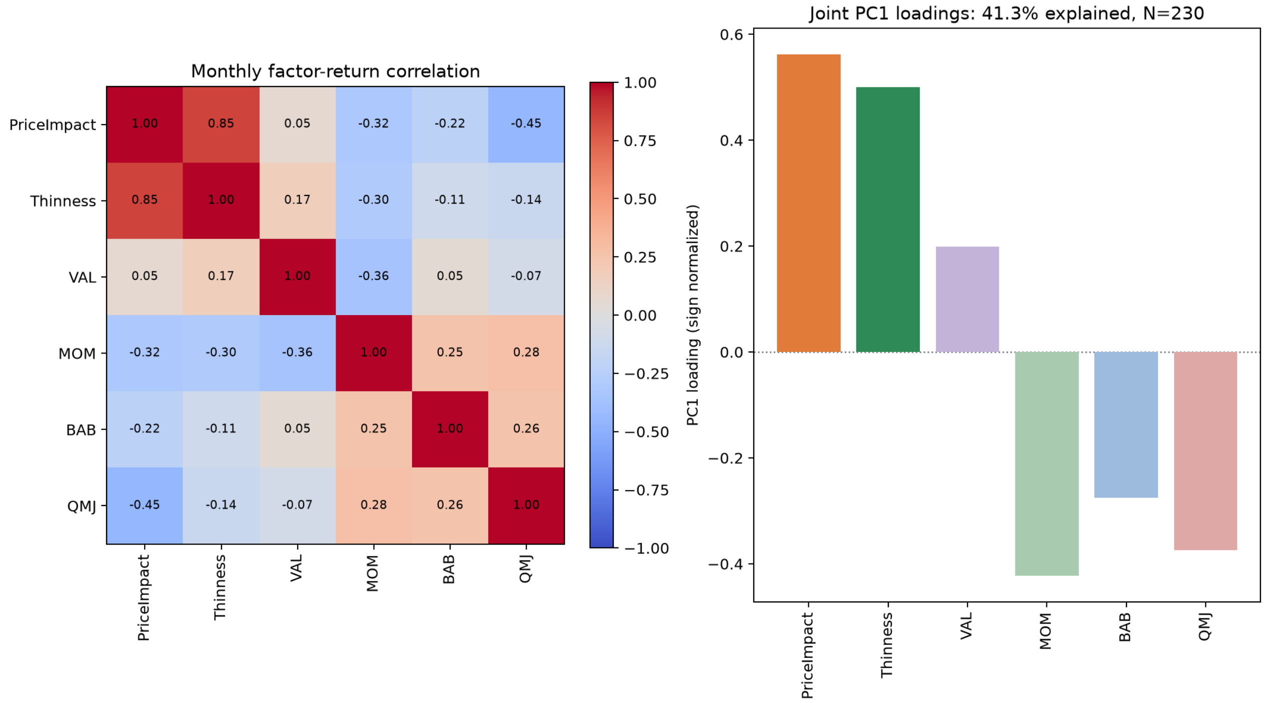

Figure 5: Monthly return correlations and sign-normalized first-principal-component loadings across the standalone sleeves and the supplied standard factors.

6 Signal persistence

Mean rank IC between formation exposure and a single later month's return. Flat profiles persist; fast decay implies faster rebalancing.

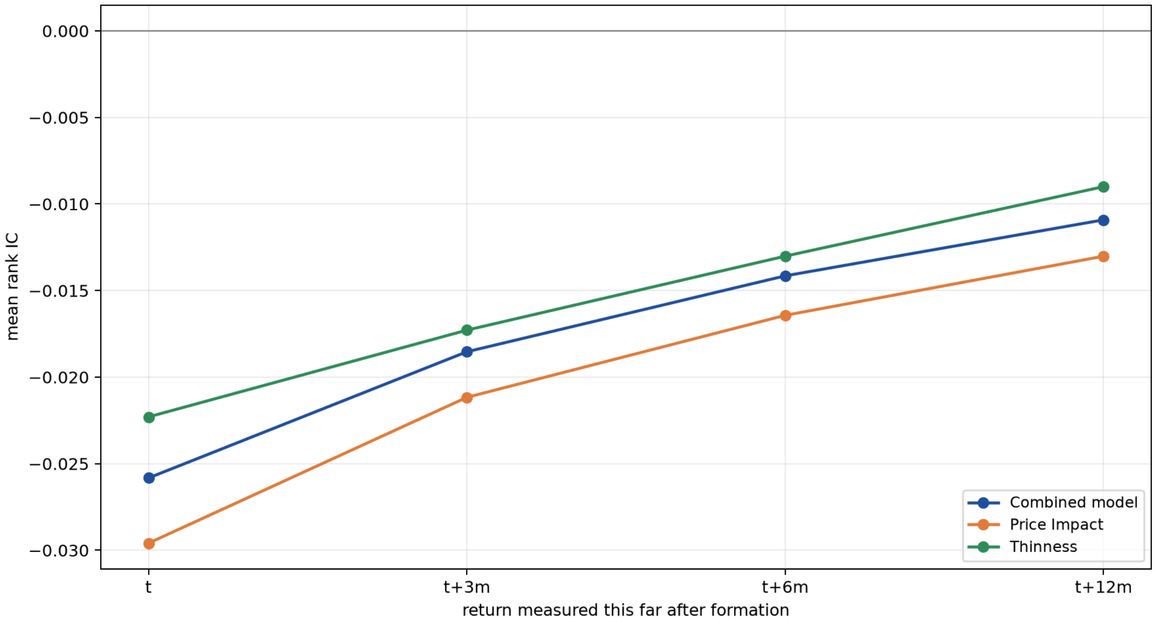

Figure 6: Mean rank information coefficient at each horizon. A line that stays flat is a signal that keeps working long after the month it is traded.

Portfolio	Horizon	Mean IC	t	Months
Combined model	t	-0.0258	-4.18	252
Combined model	t+3m	-0.0185	-3.18	248
Combined model	t+6m	-0.0142	-2.29	245
Combined model	t+12m	-0.0109	-1.82	238
Price Impact	t	-0.0296	-4.56	252
Price Impact	t+3m	-0.0212	-3.48	248
Price Impact	t+6m	-0.0164	-2.58	245
Price Impact	t+12m	-0.0130	-2.11	238
Thinness	t	-0.0223	-3.75	252
Thinness	t+3m	-0.0173	-2.88	248
Thinness	t+6m	-0.0130	-2.09	245
Thinness	t+12m	-0.0090	-1.51	239

7 Candidate comparisons and replacement

Standalone compares candidate with incumbent. In-model replaces the incumbent sleeve and rebuilds the standard model; diversification can reverse the standalone ranking.

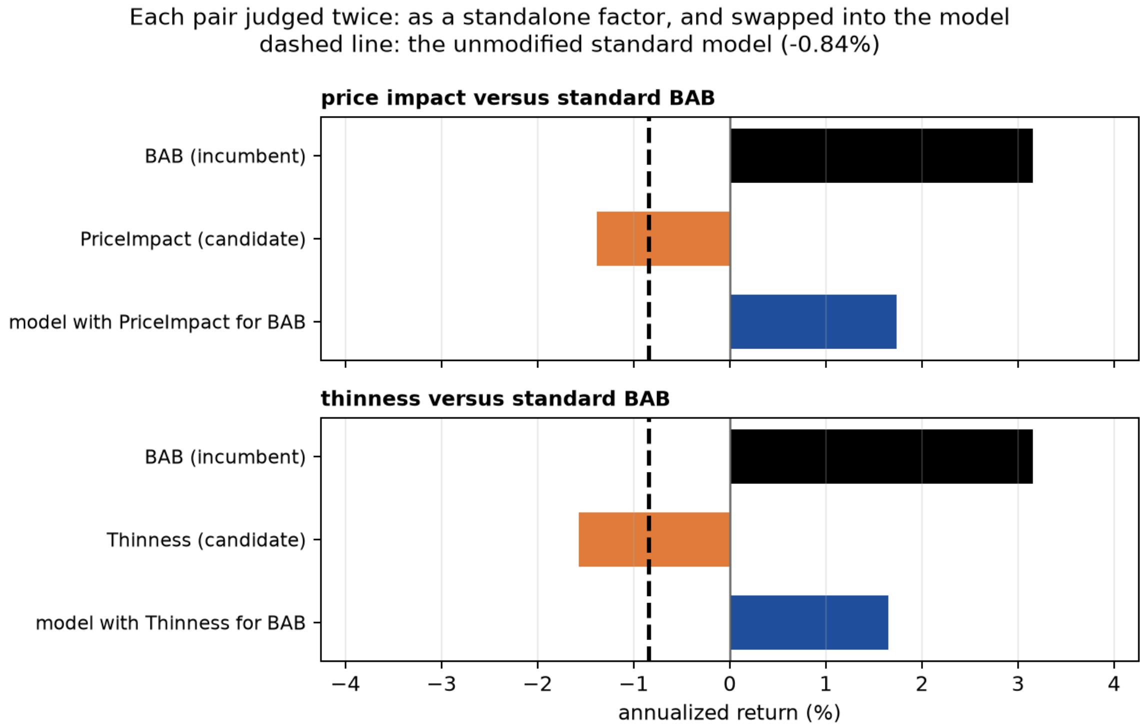

Figure 7: Each pair judged twice: the incumbent factor against the candidate standing alone, then the model with the candidate swapped in. The dashed line is the unmodified standard model.

Pair	Role	Portfolio	Ann. ret.	Change	Alpha	Change	t
price impact versus standard BAB	incumbent	BAB	3.15%	–	3.22%	–	1.03
	candidate	Price Impact	-1.38%	-4.54%	-2.32%	-5.54%	-1.31
	in model	Standard model, Price Impact for BAB	1.74%	+2.58%	3.46%	-2.33%	1.88
thinness versus standard BAB	incumbent	BAB	3.15%	–	3.22%	–	1.03
	candidate	Thinness	-1.57%	-4.73%	-0.92%	-4.14%	-0.67
	in model	Standard model, Thinness for BAB	1.65%	+2.50%	4.19%	-1.60%	2.08

8 Conclusion

Verdict. Reject for now: it clears neither control at |t| >= 2. vs SPY market benchmark: alpha -2.63%, t=-1.64; vs VAL+MOM+BAB+QMJ: alpha 0.46%, t=0.34.

9 Reproducibility

Run ID:

`20260826T140716Z_illiquidity-premium_ca43e4df9c`

Configuration hash: `ca43e4df9c`. Gold version: `version=20260826T131322Z`. The run directory contains configuration, data products, figures, source, log, and PDF. Re-runs never overwrite it.